

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Autumn-2025

Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline Course Code: MDP-3604

Total Marks-50

Time: 2.5 Hours

[Answer the following questions; Figures in the right hand margin indicate full marks]

1. How does a firm belief in the Akhirat (Hereafter life) accelerate people's moral development and consequently ensure social security? 10
2. a) Discuss the integrated impact of Fasting on Society 6
b) Mark the terrible consequences of not praying Salah 4
3. a) What do you mean by "Nutfah" and "Sulalah"? 2
b) Describe the stages of fetal development in the womb in the light of the Quran and modern embryology. 8
4. What do you understand by Miracle (Mojeza)? *"Different messengers have different types of Miracle"*- explain any four miracles from them. 10
OR
What are the negative impacts of having sex beyond of marriage, free mixing, extramarital affair, homosexuality, pornography and short dress on society? 10
5. a) Classify disasters with example 2
b) Mention two examples of natural disasters described in the Qur'an 2
c) According to Islamic approach, how to maintain good health 6
OR
5. How would you evaluate the role of private university students in Bangladesh in the July Movement? What kind of educational environment do you expect in Bangladeshi universities after the July Movement? 10

International Islamic University Chittagong
Center for General Education (CGED)

Semester End Examination: Autumn 2025
 Course Code: GEHE-3601

Program: Undergraduate
 Course Title: History of Emergence of
 Bangladesh
 Full Marks: 50

Time: 2 hours and 30 minutes.

Instructions:

- All Questions are Compulsory.
- Figures in the right margin indicate full marks.
- Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Letter of Symbol Meaning	Bloom's Levels of the Questions.					
	R Remember	U Understand	App Apply	An Analyze	E Evaluate	C Create

Text of the Questions		Marks	Bloom's Level	CLO
1	Evaluate the historical context of the formation of the United Front in 1954 highlighting major points of its election manifesto.	10	E	CLO2
2	Evaluate the trend and nature of the students' movement of 1962 against General Ayub Khan and estimate its impact on the subsequent Bangali nationalism movements.	10	An	CLO3
3	Review the causes and consequences of the Mass-Upsurge in 1969. How did this movement contribute to shaping contemporary mass uprisings in Bangladesh?	10	An	CLO3
4	Explain the factors and disparities that disintegrated Pakistan and led to the emergence of independent Bangladesh in 1971. Or Review the spontaneous resistance and sector-based strategic battle during the nine-month-long historic Liberation War of Bangladesh in 1971.	10	E	CLO3
5	Write short notes on any two: a) Legal Framework Order in 1970 b) Fundamental Rights of the Citizens c) Basic Features of the Constitution of Bangladesh.	5 5	U	CLO3

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Autumn -2025

Course Code: URED-3604(URED-3102 for LLB)
Course Title: Life and Teachings of the Prophet (SAAS)

Time: 2 Hours 30 Minutes

Full Marks: 50

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	a. Define the term <i>Hijrah</i> literally and terminologically discussing the background of the <i>Hijrah</i> of Prophet Muhammad (SAAS) from Makka to Medina.	5		Remember
	b. Illustrate the <i>Hijrah</i> of Prophet Muhammad (SAAS) as a turning point in the history of Islam.	5	3	Understand
2	Analyze the cause of the battle of Badar mentioning the key events and lessons derived from it.	10		
	Or		3	Analyze
3	Examine the background of Battle of Uhud relating the major events and lesson of it for Muslim Community.			
	Evaluate the treaty of <i>Hudaibiyyah</i> justifying why it is termed as <i>Fathum Mubeen</i> (clear victory) in the Holy Qur'an.	10	3	Evaluate
4	Explain the background of the conquest of Makkah and describe how Makkah was conquered without any bloodshed.	10	3	Create
5	Critically relate the Farewell sermon of Prophet Muhammad (SAAS) as a foundational charter of human of human rights in the history of mankind.	10	3	Apply

Right of women
 Reverse the law
 be removed from there
 Right of life
 property
 of credit & tribal society
 human

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
 Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
 Course Code: CSE-3641 Course Title: Software Engineering
 Time: 2 hours 30 minutes Total marks: 50

[Answer *all* the questions. Figures in the right-hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A			
Q. No.	Question	CL O	DL Mar ks
✓ 1.a	A hospital information system is being proposed, but doctors and nurses are unsure about the exact features they need. How would requirement elicitation and analysis help in resolving this uncertainty?	2	U 3
1.b	In a library management system, librarians, students, and IT staff provide different requirements. Identify the stakeholders and classify them into primary, secondary, and tertiary stakeholders.	3	U 4
1.c	During validation, several requirements are found to be ambiguous and not testable. Which requirement checking criteria are violated and how can they be corrected.	2	A 3
OR			
✓ 1.a	Identify the primary, secondary, and tertiary stakeholders for the university LMS project.	2	U 3
✓ 1.b	Explain the roles and interests of each category of stakeholders in the project.	3	U 4
✓ 1.c	Describe how you would engage each group of stakeholders to ensure the project's success.	2	A 3

2. **Read the passage and answer questions 2(a):**

A startup is developing a Campus Food Delivery App for university students. The system includes modules for user registration, restaurant listing, menu browsing, cart management, order placement, payment processing, and delivery tracking. The development team is divided: one group prefers starting with high-level system architecture, while another wants to begin by building and testing individual components first.

- 2.a Compare the Top-Down and Bottom-Up approaches to software design in the context of this project. Explain how each approach would be applied, and state one advantage and one challenge of each. 1 A 5
- 2.b A software company based in Bangladesh is currently facing significant challenges in three critical areas: timely and reliable product delivery, efficient utilization and management of physical and digital workspace, and frequent, severe security breaches compromising sensitive data and system integrity. In the context of software engineering requirements, identify and discuss the specific category (or categories) of requirements that address these types of organizational and operational concerns. Clearly define this requirement category, explain its characteristics, and justify why the mentioned issues fall under it rather than other types of software requirements (such as functional or user requirements). 3 E 5

3+4

Group B

Q. No.	Question	CL O	D L E	Marks
3.a	Mr. X, a software developer at a mid-sized tech firm in Dhaka, recently released a pre-final version of a mobile banking application to a group of external end-users as part of a field trial. During this phase, one of the participants entered valid transaction details into the app, but instead of <u>processing the correct amount</u>, the system displayed an <u>incorrect balance deduction</u>. Concerned about reliability and correctness, Mr. X now needs to investigate and resolve this issue before the official launch. In this context: - Briefly define what beta testing is and its role in the software testing lifecycle. - Identify and explain the general testing approaches or strategies Mr. X should employ to diagnose, isolate, and fix the erroneous behavior reported during beta testing. Justify why these approaches are appropriate for addressing such runtime output discrepancies.	4	E	5
3.b	You are part of a quality assurance team at a software company in Bangladesh developing a web-based e-learning platform that includes features such as <u>user registration</u>, <u>course enrollment</u>, <u>real-time quiz submission</u>, <u>grade calculation</u>, and <u>personalized dashboard analytics</u>. As the testing phase approaches, your team must decide how to allocate testing efforts between black-box and white-box techniques to ensure both functional correctness and internal code reliability. In this context: - Identify and describe specific scenarios or functional features of the e-learning platform that are best suited for black-box testing, and justify why a black-box approach is appropriate for each. - Similarly, identify and explain specific internal code structures, logic paths, or implementation-level concerns that should be prioritized for white-box testing, along with the rationale for choosing white-box techniques in those cases.	3	A	5
4.a	A well-established fintech startup in Bangladesh has been maintaining a legacy loan-processing system originally developed over a decade ago using outdated technologies and monolithic architecture. Recently, the company has encountered frequent system crashes, difficulty integrating with modern APIs, and an inability to scale during peak user demand. The management has decided not to build a completely new system from scratch due to budget constraints and the risk of losing critical business logic embedded in the legacy code. Instead, they have opted to re-engineer the existing software to improve its structure, performance, and maintainability while preserving its core functionality. In this context: - Elaborate on the step-by-step process of software re-engineering, including key activities such as <u>reverse engineering</u>, <u>code restructuring</u>, <u>data re-engineering</u>, and forward engineering. - Discuss the major benefits that the fintech company can expect from undertaking software re-engineering, particularly in terms of cost savings, improved reliability, enhanced maintainability, and extended software lifespan.	2	A	5

- 4.b The fan in and fan out of module X is 3 and 4 respectively. The complexity of the system is 2000. The number of lines in module X is 200, i.e LOC of module X is $LOC(X) = 200$. Calculate the structural complexity of module X using card and glass's system complexity. Using combined Henry Kafura's approach and Card glass's approach calculate the data complexity of module X. 2 A 5
- 5.a What are the key factors that influence software cost estimation? Explain the differences between empirical, heuristic, and analytical estimation techniques. 3 A 4
- 5.b Calculate the effort and time required for the three modes (organic, semi-detached, embedded) to complete a 300 KDLOC project using the basic COCOMO model 3 U 6

OR

5. Read the passage and answer questions 5(a-c):

You are the Director of Quality Assurance at a regional hospital network that has recently adopted an Electronic Health Record (EHR) system. The organization currently operates at CMM Level 2, with consistent project delivery for basic system rollouts. However, there is inconsistency in how clinical workflows are integrated across departments, limited reuse of process assets, and no formalized organizational standards for change management or training.

The Chief Medical Officer has mandated a move to CMM Level 3 within 18 months to ensure patient safety, regulatory compliance, and interoperability across all facilities.

- 5.a Outline the core organizational practices you would institutionalize to transition from CMM Level 2 to Level 3, focusing on standardization and documentation. 2 U 3
- 5.b Identify two major resistance points you anticipate during this transition (staff, systems, culture), and propose specific mitigation strategies for each. 2 A 4
- 5.c Explain how achieving CMM Level 3 will directly improve clinical outcomes and enhance stakeholder (patients, clinicians, administrators) confidence in the EHR system. 2 A 3

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 International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
Course Code: CSE3631 Course Title: Operating System
 Total marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A		CLO	BT	Marks
Q. No.	Question			
1.a	Consider the following resource allocation state of a system with four processes, P1, P2, P3, and P4, and five types of resources, RS1, RS2, RS3, RS4, and RS5 where C: the current allocation matrix, R: the request matrix. E: the existing resource vector A: the available resource vector $C = \begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 & 0 \end{bmatrix}$ $R = \begin{bmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 3 & 1 \\ 0 & 2 & 1 & 1 & 0 \end{bmatrix}$ $E = (24144)$ $A = (01021)$ Using the deadlock detection algorithm, show that there is a deadlock in the system. Identify the processes that are deadlocked.	2	C5	5
1.b	Determine whether in the following scenario, represented by Resource Allocation Graph, deadlock exists or not.	2	C5	5
2.a	Consider a memory partition with the following hole sizes in order: 8 KB, 20 KB, 15 KB, 30 KB, and 45 KB. i. Which holes are taken for successive segment requests of 12 KB, 10 KB, and 25 KB for First Fit and Worst Fit? ii. Explain the "External Fragmentation" trade-off between Best Fit and Worst Fit based on this sequence.	3	C5	5
2.b	Describe the function of the Page Table and the Memory Management Unit (MMU). Why is a Translation Lookaside Buffer (TLB) necessary in a paging system?	3	C5	5
OR				
2.a(or)	An operating system environment has five free memory blocks with sizes 120 KB, 300 KB, 60 KB, 200 KB, and 500 KB. Five new programs arrive at the system, each hungry for memory: Program A needs 112 KB, Program B needs 250 KB, Program C needs 50 KB, Program D needs 425 KB, and Program E needs 95 KB. Using the First Fit, Best Fit, and Worst Fit memory allocation strategies, determine how each program will be allocated, show the state of memory blocks after all allocations, identify any programs that remain unallocated, and analyze which strategy leads to the least external fragmentation in this scenario.	3	C5	5
2.b(or)	A system supports multiprogramming and has limited main memory. Several processes are waiting to execute, but the total memory required exceeds the available memory. Explain how swapping can be used to manage memory in this situation. Discuss the steps involved	3	C5	5

	in swapping a process out of main memory to secondary storage and then bringing it back, the advantages and disadvantages of using swapping, and how it affects CPU utilization and process execution time.			
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Group B

Q. No.	Question	CLO	BT	Mark
2.a	Discuss the concept of "demand paging" in virtual memory systems. How to Steps in Handling a Page Fault occurs. Explain with proper diagram.	3	C5	5
2.b	Consider the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2. Assuming a physical memory with three (3) frames that are initially empty, illustrate the frame allocation for the following algorithms: i. LRU (Least Recently Used) ii. Optimal Page Replacement For each algorithm, determine the number of Page Faults and calculate the Hit Ratio.	3	C5	5
4.a	A disk drive has 200 cylinders (numbered 0 to 199). The current position of the read-write head is at cylinder 50. The queue of pending requests, in FIFO order, is: 82, 170, 43, 140, 24, 16, 190. Assuming the head is moving towards the higher cylinder numbers, calculate the total head movement (seek distance) using: i. C-SCAN (Circular SCAN) ii. C-LOOK (Circular LOOK)	3	C5	5
4.b	In an operating system, files can be accessed in different ways depending on the organization and intended use. Explain the various file access methods including sequential access, direct (random) access, and indexed access. Discuss the advantages and disadvantages of each method, provide examples of scenarios where each method is suitable and analyze how the choice of access method affects file system performance and efficiency.	3	C5	5
OR				
4.a (or)	In contemporary storage subsystems, efficient I/O request arbitration is paramount for optimizing disk throughput and minimizing average data access latency. Consider a direct-access storage device (DASD) with 300 physical tracks (numbered 0 to 299). A persistent pending I/O queue contains requests for data residing at the following track addresses: 115, 23, 187, 85, 10, 163, 68. The Read/Write head's current operational locus is at Track 75, and its initial traverse direction is towards higher track numbers. Assume a linear seek time model where each track movement incurs a latency of 0.5 ms. Systematically, execute the SCAN and C-SCAN disk scheduling algorithm for the given I/O request queue and perform a critical comparative analysis between them based on the results and understanding of their operational principles.	3	C5	5
4.b (or)	In an operating system, files can be accessed in different ways depending on the organization and intended use. Explain the various file access methods including sequential access, direct (random) access, and indexed access. Discuss the advantages and disadvantages of each method, provide examples of scenarios where each method is suitable and analyze how the choice of access method affects file system performance and efficiency.	3	C5	5
5.a	What commonly used computer programs are prone to man-in-the-middle attacks? Discuss solutions for preventing this form of attack.	3	C5	5
5.b	What is the significance of the "Switch" right in an Access Matrix? How does it allow a process to change its protection domain? Explain with a proper example.	3	C5	5

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
Course Code: CSE3525 Course Title: Data Communication
 Total marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Why is Quadrature Amplitude Modulation (QAM) considered the most spectrally efficient digital-to-analog modulation technique- <u>explain it clearly</u> with the limitations of ASK, FSK, and PSK? Also draw the <u>constellation diagram</u> for OOK and 16-QAM.	2	U	5
1.b	Find the bandwidth of a signal transmitting at 12 Mbps for QPSK. Assume the value of roll off factor = 1. Explain the generation of a QPSK signal for the bit sequence 00100111 with the help of a neat labelled waveform diagram?	2	A	5
OR				
1.a	What do you mean by signal rate and data rate? Write the relationship between signal rate and data rate. An analog signal has a bit rate of 16000 bps and a band rate of 1000 baud.	2	U	5
	i) How many data elements are carried by each signal element? ii) How many signal elements do we need?			
1.b	A corporation possesses a satellite channel with a bandwidth of 10 MHz. The corporation intends to establish 40 distinct and independent channels, each capable of transmitting a minimum of 2 Mbps. Design a suitable modulation technique for this communication requirement.	2	A	5
2.a	What is the key advantage of Statistical TDM over Synchronous TDM? Four 1-kbps connections are need to be combined using Synchronous Time Division Multiplexing with a unit of 1 bit. Answer the following questions regarding the multiplexing process: i. What is the duration of a time slot? ii. What is the transmission rate of the link? iii. What is the duration of 1 bit before multiplexing? iv. What is the duration of a frame?	3	A	5
2.b	What is a guard band in FDM and why is it needed? Four channels, each with a bandwidth of 100 kHz, are to be multiplexed together. If a guard band of 10 kHz is required between adjacent channels, calculate the minimum bandwidth of the link? Also, briefly explain one real-world application of FDM where guard bands are important?	3	An	5

Group B

Q. No.	Question	CLO	BT	Marks
3.a	Explain bit stuffing using the data 011111010111110 and show the resulting stuffed bit stream? Also, briefly describe the disadvantages of Stop-and-Wait protocol, including the effects of frame or ACK loss during transmission?	2	A	5
3.b	Explain the key advantage of selective repeat over Go-Back-N protocol? Assume a sender sends 6 packets numbered 0, 1, 2, 3, 4, 5. The sender receives an acknowledgment with ack_no = 3. Explain the interpretation of this acknowledgment in the following protocols: i. Go-Back-N (GBN)	2	A	5

ii. Selective Repeat (SR)

4.a Define Burst-Error? Suppose you have a 4-bit data-word 1001 and you are using Hamming Code C(7,4). Show the codeword at the sender site. During transmission, a single-bit error changes the received data-word to 1101. Show the received codeword and explain how the receiver identifies the position of the erroneous bit?

3 A 5

4.b Explain the key disadvantage of a parity bit with an appropriate example? Using CRC, if the data word represented by the polynomial $D(x) = x^6 + x^4$ and the generator polynomial is $G(x) = x^3 + x + 1$. Determine the codeword and show how the receiver verifies it using modulo-2 division?

3 A 5

OR
4.a Explain Single-Bit Error and Burst-Error. How Two-Dimensional Parity Check is performed?

3 A 5

4.b Suppose the sender has encoded the dataword 1001 using Cyclic Redundancy Check (CRC) where the divisor is from the polynomial of $x^3 + x + 1$. The receiver receives the codeword 1000110. Explain whether the receiver receives the correct codeword or not by using CRC decoder for receiver side.

3 A 5

5.a An Ethernet frame contains the source MAC address 43:6A:7C:1D:10:00, but the receiver discards the frame. Explain why the frame is discarded? Also, classify the following destination MAC addresses as unicast, multicast, or broadcast, giving reasons for each:

4 U 5

- 4A:30:10:21:10:1A
- FF:FF:FF:FF:FF:FF
- 47:20:1B:2E:08:EE

5.b Explain the hidden node problem and the exposed node problem in Wireless LAN (IEEE 802.11)? Draw suitable diagrams to illustrate each case. Also briefly explain one mechanism used in IEEE 802.11 to reduce these problems?

4 U 5

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group-A

1. a)



Consider the following sentences in the Wumpus world above:

- R1: $\neg P_{1,1}$
 R2: $B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$
 R3: $B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 R4: $\neg B_{1,1}$
 R5: $B_{2,1}$

KB containing R1 through R5, and show how to prove $P_{3,1}$ that is, there is pit in [3,1].

CLO Marks
 CLO2 6

1. b) Consider the following axioms:

- P (1)
 $(P \wedge Q) \rightarrow R$ (2)
 $(S \vee T) \rightarrow Q$ (3)
 T (4)

- a) Convert the formulas into CNF.
 b) Prove that whether R is true or not using any inference technique.

CLO2 4

OR

1. a) Convert the following sentences to propositional logic:

- i) If it is morning, then the birds are singing.
 ii) I am eating if and only if I am hungry.

CLO2 1+1

1. b) Consider the following statements-

- John likes all kind of food.
- Apple and vegetable are food.
- Anything anyone eats and not killed is food.
- Anil eats peanuts and still alive.
- Harry eats everything that Anil eats.

CLO2 6

Using the Resolution technique, prove the statement "John Likes Peanuts".

1. c) Discuss two differences between forward chaining and backward chaining.

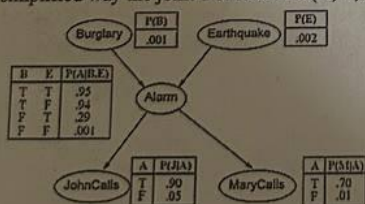
CLO2 2

2. a) In a medical diagnosis system, the probability that a patient has Disease D is 0.002. The probability that a patient shows symptom S given the disease is 0.95, and the probability of showing the symptom without the disease is 0.04. Calculate the probability that the patient actually has the disease given that the symptom is observed.

CLO3 4

2. b) A Bayesian network showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm, John Calls, and Mary Calls respectively. The independent conditional probability helps us to write in a simplified way the joint distribution $P(H, E, A, I, M)$

CLO3 6



i) Express the joint distribution $P(B, E, A, J, M)$ in terms of the conditional probabilities (and independences) expressed in the Bayesian Network above.

ii) Probability of the event that the alarm has sounded but neither a burglary nor an earthquake has occurred and both Mary and John call.

0.1/27

iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred, and both Mary and John call.

Group-B

3. Explain Bayes Theorem. Clearly define prior probability, likelihood, and posterior probability. CLO3 3
 3. Given the following membership function for the fuzzy set "Warm Temperature": CLO3 4

$$\mu_{\text{warm}}(x) = \begin{cases} 0, & x \leq 15 \\ \frac{x-15}{10}, & 15 < x \leq 25 \\ 1, & x > 25 \end{cases}$$

- (a) Explain what a membership function is in fuzzy logic.
 (b) Find the degree of membership of a temperature of 20°C in the fuzzy set "Warm". CLO3 3
 3. In a city, 60% of drivers use mobile phones while driving. The probability of an accident when a driver uses a phone is 0.05, and when not using a phone is 0.01. If an accident has occurred, calculate the probability that the driver was using a mobile phone. CLO3 4
 4. Contrast expert systems and neural networks in terms of knowledge representation, knowledge acquisition and explanation. Give one domain in which the expert system approach would be more promising and one domain in which neural network approach would be more promising. CLO3 6
 4. A software company is developing an AI-powered customer support chatbot for an online shopping platform. The chatbot must understand customer messages, track conversation context, and respond politely even when customers are indirect or emotional.

Answer the following:

- i. Explain Natural Language Processing (NLP) and justify why NLP is necessary for this chatbot.
 ii. Write down both the top down and bottom up parse tree for the sentence:
 "The chatbot resolved the issue".

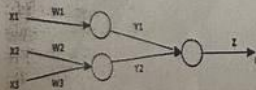
OR Using the following simple grammar for a fragment of English.

S → NP VP
 NP → Det N
 VP → V NP
 Det → the
 N → chatbot
 N → issue
 V → resolved
 V → handled
 NP → N
 VP → V

5. a) A neuron in an ANN uses a sigmoid activation function: CLO4 2+1+1
 $f(x) = 1 / (1 + \exp(-x))$

- a) Name one similarity and one difference between artificial and biological neural networks.
 b) Explain how the sigmoid function helps in deciding between two classes in binary classification.

- c) If the input to the neuron is $x = 1$, find the sigmoid output. CLO4 2+2+2
 5. b) Review the following neural network and compute the output Z.



The input values are $X_1=15$, $X_2=8$, $X_3=14$ and the corresponding weights are: $W_1=0.6$, $W_2=0.3$, $W_3=0.1$, $Y_1=0.6$, $Y_2=0.45$.

- a) Compute the value of Z without using any transfer (activation) function.
 b) Apply a threshold at all neurons. If the value is 5 or less, output 0; otherwise, output 1. Recalculate Z.
 c) Compute the value of Z assuming that the ReLU transfer function is used at all neurons.

- OR
 5. a) What is semi-supervised learning? Explain the concept gradient descent optimization. CLO4 1+3
 5. b) Consider a single neuron with two inputs X_1 and X_2 , and output Y, learning the logical AND operation. The truth table is given below. CLO4 6

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Develop an ANN model using Delta rule, Where,
 Threshold, $T = 0.5$; Learning rate, $\alpha = 0.2$; Initial weights $W_1=0.1$ and $W_2=-0.5$

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE
Final Exam, Autumn 2025

Course Code: **ECON 3501**

Time: 2 hours 30 minutes

Course Title: **Principle of Economics**
Full Marks: 50

- (i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions		PO	BL
CO	CO Statements		
CO1	Explain the knowledge of the fundamental concepts and theories of micro and macro-economics.	PO11	C1,C2
CO2	Analyze the key indicators of economic growth.	PO11	C4
CO3	Compare the economic theories and concepts to analyze behavior of individuals, firms and nations to act as a responsible citizen.	PO11	C5

Bloom's Levels (BL) of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part-A: Answer the following questions

- Q1 a Define market in economics and explain why a market need not be a physical place? CO2 R 2
 b Explain the concept of market structure and discuss its main types. CO2 An 5
 c Distinguish between monopoly and monopolistic competition market. CO2 R 3
- OR**
- a How does the oligopoly market differs from perfect competitive market and What do you mean by the term "interdependence"? CO2 R 5
 b Explain the relationship between AC (average cost) and MC (marginal cost) and why the MC curve is U-shaped? CO2 An 5
- Q2 a Define inflation and mention reasons behind inflation. CO2 E 3
 b Show the differences between GDP and GNP. CO2 An 4
 c The following is information from the national income accounts for a hypothetical country: GDP = \$6,000, Gross Investment = \$800; Net Investment = \$200, Consumption = \$4,000, Government Purchases of Goods and Services = \$1,100 and Government Budget Surplus = \$30. What is (i) NDP? (ii) NX? (iii) DPI? CO2 C 3

Part-B: Answer the following questions

- Q3 a Differentiate between expansionary monetary policy and contractionary monetary policy. CO3 An 2
 b Explain the working mechanism of monetary policy instruments with reference to Bangladesh. CO3 E 4
 c Identify types of trade policy and explain their basic objectives. CO3 An 4

- Q4 a) Show the differences between economic growth and economic development. CO3 An 5
 Explain the Harrod-Doman Growth Model. CO3 E 5
 b) Define national growth rate and show how it is calculated. Show how the relationship between population growth and technological progress contribute in economic growth.
- Q5 a) Differentiate between functional planning and structural planning. Discuss CO4 E 5
 the objectives of planning. CO4 An 5
 b) Explain Linear Model and Stages Model of policy making. CO4 E 5
OR
 What are the factors that determine economic growth of a country? CO4 E 5
 How can a country with a large population but scarce natural resources (e.g., Japan, Singapore) achieve high levels of development? CO4 An 5

making
 selection
 natural

growth
 development
 natural

Q4 a) Show the differences between economic growth and economic development. CO3 An 5
 Explain the Harrod-Doman Growth Model. CO3 E 5
 b) Define national growth rate and show how it is calculated. Show how the relationship between population growth and technological progress contribute in economic growth.